

Teacher's Signature: N/A

Date: 10-12-25

United International University

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Section: X Group: 04 Trimester: Fall 2025 Date: 10-12-25

Experiment No. 06

6. a : Name of the Experiment : To investigate the characteristics of a series DC circuit and to verify Kirchoff's Voltage Law (KVL).

OBJECTIVE:

The objective of this experiment is to investigate the characteristics of a series DC circuit and to verify Kirchoff's Voltage Law (KVL).

THEORY:

In a series circuit (Figure 3.1) the current is same through all of the circuit elements.

The equivalent Resistance, $R_T = R_1 + R_2 + R_3$.

By Ohm's Law, the Current is

$$I = \frac{V_{supply}}{R_T}$$

KVL states that the voltage rises must be equal to the voltage drops around a close circuit. Applying Kirchoff's Voltage Law around closed loop of Figure 3.1, we find,

$$V_{supply} = V_1 + V_2 + V_3$$

Where, $V_1 = IR_1$, $V_2 = IR_2$, $V_3 = IR_3$

Current I is same throughout the circuit for figure 3.1.

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The voltage divider rule states that the voltage across an element or across a series combination of elements in a series circuit is equal to the resistance of the element divided by total resistance of the series circuit and multiplied by the total impressed voltage. For the elements of Figure 3.1

$$V_3 = \frac{R_3 E}{R_T}$$

$$V_1 = \frac{R_1 E}{R_T},$$

$$V_2 = \frac{R_2 E}{R_T},$$

EQUIPMENTS:

- Variable DC power supply -1piece
- Digital Multimeter (DMM)/ Analog multimeter-1piece.
- Resistances: 100 Ω , 220 Ω , 470 Ω -1piece each.
- Trainer Board-1piece
- Connecting Wires.

CIRCUIT DIAGRAM:

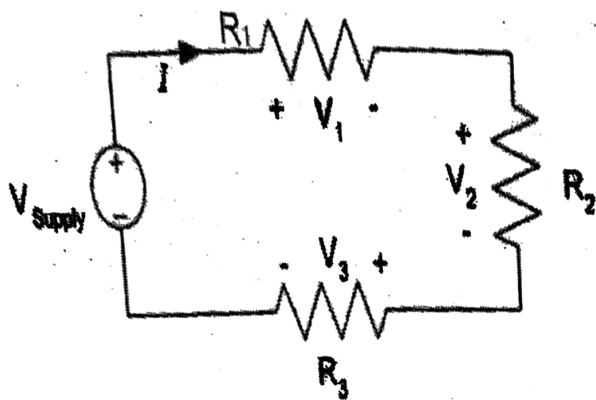


Figure 3.1

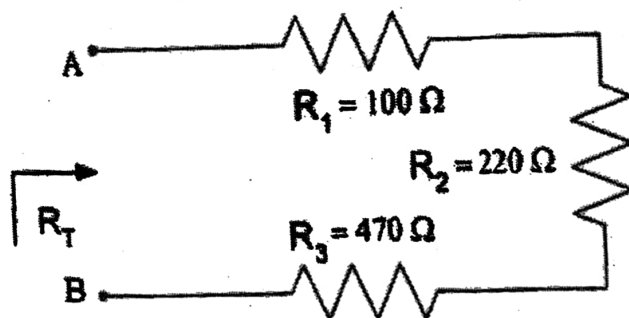
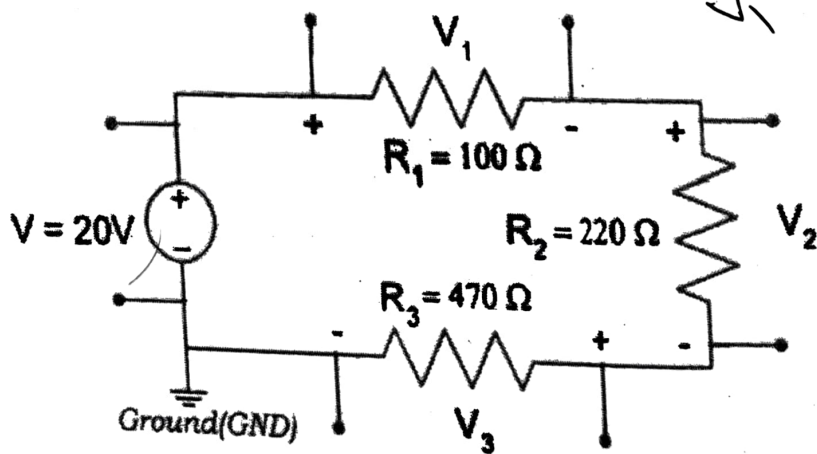


Figure 3.2



Handwritten calculation:

$$V_1 = \frac{98}{770} \cdot 20$$

Below the equation, there are handwritten notes: "V2 = V1" and "V3 = V1".

Figure 3.3

PROCEDURE:

1. Measure the resistances having values $100\ \Omega$, $220\ \Omega$ & $470\ \Omega$ by using Ohmmeter and record the values in Table 3.1
2. Construct the circuit as shown in Fig 3.2.
3. Then measure input resistance R_T across points A-B using Ohmmeter and record that value in Table 3.1.
4. Now construct the circuit as shown in Fig 3.3. Turn on the DC power supply and set the DC supply to 20V by using Voltmeter.
5. Measure voltage across each resistor with Voltmeter and record in the Table 3.1
6. Calculate V_1 , V_2 and V_3 using Voltage Divider Rule (VDR). [Use measured values of resistances for all calculations.]

Experimental Data:

(A) The supply voltage, $V_{\text{supply}} = V_s = V_{\text{in}} = V_{\text{The}} = V = \dots\dots\dots \text{Volt}$

(B) Tolerance Color = $\dots\dots\dots$ and Tolerance value as percentage = $\pm \dots\dots\%$

(C) Tolerance value = Maximum or Minimum = $\dots\dots\dots$

DATA SHEET:

Table 3.1

Nominal values of Resistance (Ω)	Measured values of Resistance by Ohmmeter (Ω)	Measured values of Resistance by Tolerance (Ω)	Equivalent Resistance, R_T		Measured voltage across each resistor (V)	Calculated Voltage using VDR (V)
			Measured R_T by using Ohmmeter (Ω)	Calculated $R_T = R_1 + R_2 + R_3$ (Ω)		
$R_1 = 100$	98		771	769	$V_1 = 2.56$	2.54
$R_2 = 220$	215				$V_2 = 5.64$	5.6
$R_3 = 470$	456				$V_3 = 11.85$	11.86

Calculation:

$$V_1 = \frac{R_1}{R_T} \cdot V_S = \frac{98}{769} \cdot 20 = 2.54 \text{ V}$$

$$V_2 = \frac{R_2}{R_T} \cdot V_S = \frac{215}{769} \cdot 20 = 5.6 \text{ V}$$

$$V_3 = \frac{R_3}{R_T} \cdot V_S = \frac{456}{769} \cdot 20 = 11.86 \text{ V}$$

$$V_1 + V_2 + V_3 = 20.05 \approx V_S$$

KVL verified

Signature of the Teacher

Discussions:

Q: What can you deduce about the characteristics of a series circuit from observation Table 3.1?

1. The same current flows through all components.
2. Total resistance of the circuit is equal to the sum of individual resistance.
3. The voltage drop across each resistor is proportioned to its resistance.

Q: From the data found in Table 3.1, mathematically prove that the current in the series network of figure 3.3 is equal for each resistance.

$$R_T = 769 \Omega$$

$$V = 20 \text{ V}$$

$$I = \frac{20}{769} = 0.026 \text{ A}$$

$$I_1 = \frac{2.6}{100} = 0.026 \text{ A}$$

$$I_2 = \frac{5.63}{215} \approx 0.026 \text{ A}$$

$$I_3 = \frac{11.86}{456} \approx 0.026 \text{ A}$$

$$I \approx I_1 \approx I_2 \approx I_3$$

So, the current is the same through each resistor.

Q: Verify KVL from the data obtained in Table 3.1.

$$\text{KVL, } V_T = V_1 + V_2 + V_3$$

$$V_T = 20$$

$$\begin{aligned} \therefore V_1 + V_2 + V_3 &= 2.56 + 3.64 + 41.85 \\ &= 20.05 \text{ V} \\ &\approx 20 \text{ V} \end{aligned}$$

$\therefore$ KVL verified

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Experiment No. 06

6. b : Name of the Experiment: To investigate the characteristics of a Parallel DC circuit and to verify Kirchoff's Current Law (KCL).

OBJECTIVE:

The objective of this experiment is to investigate the characteristics of a parallel DC circuit and to verify Kirchoff's Current Law (KCL).

THEORY:

In a parallel circuit (Figure 4.1) the voltage across parallel elements is the same.

The total or equivalent resistance (R_T) is given by,

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_N}$$

If there are only two resistors in parallel, it is more convenient to use,

$$R_T = (R_1 R_2) / (R_1 + R_2)$$

In any case, the total resistance will always be less than the resistance of the smallest resistor of the parallel network.

KCL states that the currents entering a node must be equal to the currents leaving that node. For the network of Figure 4.1, the currents are related by the following expression:

$$I_T = I_1 + I_2 + I_3 + \dots + I_N$$

Applying current divider rule (CDR) for a circuit of only two resistors in parallel as shown in figure 4.2,

$$I_1 = \frac{R_2 I_T}{R_1 + R_2} \quad \text{and} \quad I_2 = \frac{R_1 I_T}{R_1 + R_2}$$

For equal parallel resistors, the current divides equally and the total resistance is the value of one divided by the 'N' number of equal parallel resistors, i.e.:

$$R_T = \frac{R}{N}$$

For a parallel combination of N resistors, the current I_1 through R_1 is:

$$I_1 = I_T \times \frac{\frac{1}{R_1}}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_N}}$$

EQUIPMENTS:

- Variable DC power supply - 1 piece
- Digital Multimeter (DMM)/ Analog multimeter - 1 piece.
- Resistances: $1\text{ K}\Omega$, $2.2\text{ K}\Omega$, $4.7\text{ K}\Omega$ - 1 piece each.
- Trainer Board - 1 piece
- Connecting Wires.

CIRCUIT DIAGRAM:

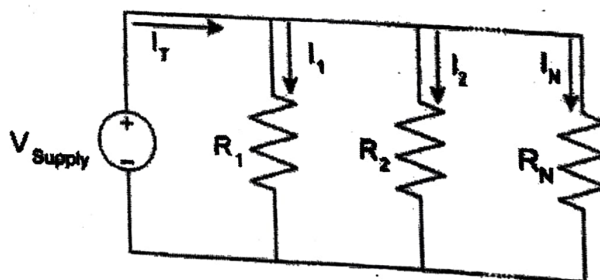


Figure: 4.1

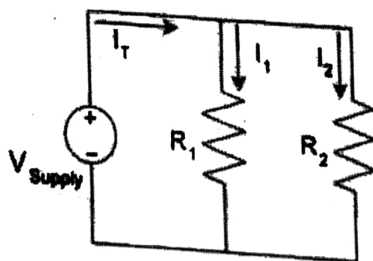


Figure: 4.2

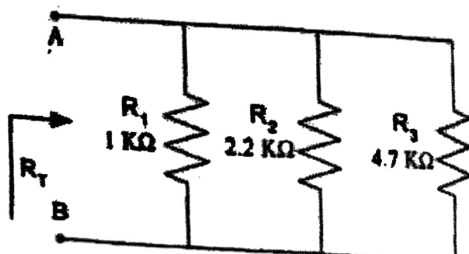


Figure: 4.3

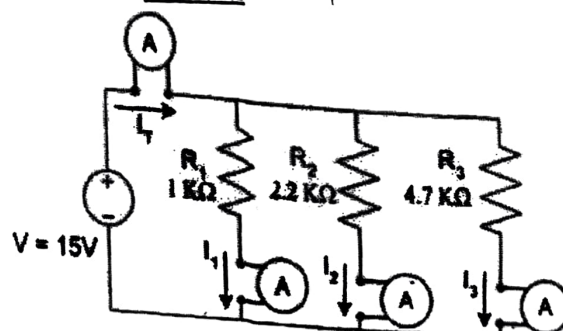


Figure: 4.4

DATA SHEET:

Table 4.1

Nominal values of Resistance ($k\Omega$)	Measured values of Resistance by Ohmmeter ($k\Omega$)	Measured values of Resistance by Tolerance ($k\Omega$)	Equivalent Resistance, R_T		Measured current through each resistor (A)	Calculated Current using CDR (A)
			Measured R_T by using Ohmmeter ($k\Omega$)	Calculated $1/R_T = 1/R_1 + 1/R_2 + 1/R_3$ ($k\Omega$)		
$R_1=1$	0.9786		0.58	0.58	$I_1=$ 14.9	15.37
$R_2=2.2$	2.19				$I_2=$ 6.4	7.04
$R_3=4.7$	4.46				$I_3=$ 3.1	3.36

Calculation:

$$I_T = \frac{V}{R_T} = \frac{15V}{0.58k\Omega} = 25.86mA$$

$$I_1 = \frac{R_T}{R_1} \cdot I_T = \frac{0.58}{0.976} \times 25.86 = 15.37mA$$

$$I_2 = \frac{0.58}{2.13} \times 25.86 = 7.04mA$$

$$I_3 = \frac{0.58}{4.46} \times 25.86 = 3.36mA$$

$$I_1 + I_2 + I_3 = 24.9 \approx I_T$$

$\therefore KCL$ verified

Signature of the Teacher

PROCEDURE:

1. Measure the resistances having values $1\text{ K}\Omega$, $22\text{ K}\Omega$ & $4.7\text{ K}\Omega$ by using Ohmmeter and record the values in Table 4.1.
2. Construct the circuit as shown in Fig 4.3.
3. Then measure input resistance R_T across points A-B using Ohmmeter and record that value in Table 4.1.
4. Now construct the circuit as shown in Fig 4.4. Turn on the DC power supply and set the DC supply to 15V by using Voltmeter.
5. Measure the currents I_T , I_1 , I_2 and I_3 by using Ammeter and record in the Table 4.1.
6. Calculate I_1 , I_2 and I_3 using Current Divider Rule (CDR). [Use measured values of resistances for all calculations.]

Experimental Data:

(A) The supply voltage, $V_{\text{supply}} = V_s = V_{\text{in}} = V_{T(s)} = V_T = V_{T\text{he}} = V = \dots\dots\dots \text{Volt}$

(B) Tolerance Color = $\dots\dots\dots$ and Tolerance value as percentage = $\pm \dots\dots\%$

(C) Tolerance value = Maximum or Minimum = $\dots\dots\dots$

Discussions:

Q: What can you deduce about the characteristics of a parallel circuit from observation Table 4.1?

1. Voltage across each resistor is the same
2. Current divides among branches.
3. Equivalent resistance is always less than the smallest resistance

Q: From the data found in Table 4.1, Calculate I_1 , I_2 , and I_3 using Ohm's Law.

$$R_1 = 0.97 \text{ k}\Omega$$

$$R_2 = 2.18 \text{ k}\Omega$$

$$R_3 = 4.46 \text{ k}\Omega$$

$$\text{Supply Voltage } V = 15\text{V}$$

$$\therefore I_1 = \frac{15}{0.97} = 15.41 \text{ mA} = 15.11 \text{ mA}$$

$$\therefore I_2 = \frac{15}{2.18} = 6.88 \text{ mA} = 7.04 \text{ mA}$$

$$\therefore I_3 = \frac{15}{4.46} = 3.36 \text{ mA}$$

Q: Verify KCL from the data obtained in Table 4.1.

KCL, law,

$$I_T = I_1 + I_2 + I_3$$

$$I_T = 25.5 \text{ mA}$$

$$I_1 + I_2 + I_3 = 24.4$$

$\therefore$ KCL verified

(A)